

sdmTMB estimation of PIBKC mature male biomass from the NMFS EBS survey

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0.1 Introduction

As noted in the previous section, the NMFS EBS bottom trawl survey captured no mature male PIBKC in 2023 and 2024, introducing problematic zeros into the survey MMB time series that the `rema` model is used to smooth. It was suggested that making assumptions regarding the sampling distribution for mature males was a simple way to develop “reasonable” replacement values for the zeros when using `rema` to obtain the smoothed time series. Although probably better than filling in the zeros with an arbitrary “small” number (as suggested in the `rema` documentation) or simply dropping the zeros from the time series, the suggested method does not avoid requiring possibly arbitrary assumptions regarding the underlying sampling distribution and/or appropriate detection level. One significant drawback to this approach is that it makes the implicit assumption that all sampling locations are statistically interchangeable and independent. In fact, PIBKC mature males have undergone a range contraction over time, as well as a decline in overall abundance/biomass (e.g., Figure 1), indicating that the sampling locations are not interchangeable or independent.

Another, more satisfying, alternative for “filling in the zeros” would be to use a spatiotemporal model such as provided by the VAST (Thorson et al. 2015) or sdmTMB (Anderson et al. 2024) R packages. Either package provides the ability to statistically model and estimate trends and variability in stock biomass through space and time given spatially-explicit data such as the NMFS EBS bottom trawl survey provides. Here, I investigated the use of sdmTMB, a more user-friendly alternative to VAST, to estimate a stock biomass time series for PIBKC mature males in the context of survey sampling results that captured no crab in multiple years.

0.1.1 Methods

sdmTMB is an R package that fits spatiotemporal Generalized Linear Mixed Effects Models (GLMMs) using Template Model Builder (TMB), R-INLA, and Gaussian Markov random fields. One common application is to create species distribution models (SDMs) from species abundance data collected across space and through time, although its principal advantage here is that it can also estimate stock-level time series by integrating estimated spatially-explicit patterns of abundance or biomass across space. A simple way to think about an sdmTMB model is that it provides a way to interpolate survey observations across space and time, taking into account both population variability across space and time as well as variability inherent to the sampling process itself.

Here, four sdmTMB models (Table 1) were fit to the spatially-explicit annual survey CPUE data (biomass/unit area) for PIBKC mature males (Figure 1) using a spatial mesh defined by the sampling locations in UTM coordinates and a cutoff of 5 km (2). Two of the four models, “m3” and “m3-ar” were two-stage “delta-gamma” models that used a binomial family with a log-link to model presence/absence of mature males in a survey haul (the delta model) and a gamma family with a log-link to model non-zero CPUE given that mature males were caught in the given survey haul (the gamma model). Both models included the survey haul depth as a possible covariate, with the possible effects on presence/absence or non-zero CPUE estimated as a smoothly-varying function of $\ln(\text{depth})$. Spatiotemporal residuals were modeled as a random walk process in m3; in m3-ar, they were modeled as a first-order autoregressive process. Additionally, the spatial correlation was allowed to be anisotropic and Restricted Maximum Likelihood (REML) was used to fit both models. The remaining two models, “tw” and “tw-ar”, were similar to their respective m3 counterparts except that models used the Tweedie distribution (Tweedie 1984) with a log-link as the statistical basis for the fit rather than a delta-gamma model.

0.1.2 Results

Standard model results are presented in Tables 2-5. AIC values for the four models are compared in Table 6 and residual diagnostics using simulated residuals and the DHARAM R package (Hartig 2022) are presented in Figures 3-6. Of the four models, the “tw-ar” had the smallest AIC and REML score, while both “ar” models had lower AIC and REML scores than their corresponding “non-ar” models.

“‘ # References {-} ::: {#refs} :::

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Table 1. Brief descriptions of the four sdmTMB models.

model	type	link(s)	residuals	covariates	anisotropic?	REML?
m3	delta-gamma	log	random walk	log(bottom depth)	true	true
m3-ar	delta-gamma	log	AR(1)	log(bottom depth)	true	true
tw	Tweedie	log	random walk	log(bottom depth)	true	true
tw-ar	Tweedie	log	AR(1)	log(bottom depth)	true	true

Table 2. Summary of results for model 'm3'.

### Results for model m3—	
Spatiotemporal model fit by REML ['sdmTMB']	
Formula: wgtCPUE ~ s(log_depth)	
Mesh: mesh3 (anisotropic covariance)	
Time column: year	
Data: dfr	
Family: delta_gamma(link1 = 'log', link2 = 'log', type = 'poisson-link')	
Delta/hurdle model 1: —————	
Family: binomial(link = 'log')	
coef.est	coef.se
(Intercept)	-3.13 1.91
slog_depth	0.07 1.51
Smooth terms:	
Std. Dev.	
sds(log_depth)	4.88
Matérn anisotropic range (spatial): 192.8 to 322.9 at 156 deg.	
Spatial SD: 2.17	
Spatiotemporal RW SD: 0.72	
Delta/hurdle model 2: —————	
Family: Gamma(link = 'log')	
coef.est	coef.se
(Intercept)	-0.21 1.78
slog_depth	-0.20 0.14
Smooth terms:	
Std. Dev.	
sds(log_depth)	0
Dispersion parameter: 2.09	
Matérn anisotropic range (spatial): 485.6 to 813.2 at 156 deg.	
Spatial SD: 0.88	
Spatiotemporal RW SD: 0.09	
REML criterion at convergence: 1114.711	

Table 3. Summary of results for model 'm3-ar'.

### Results for model m3-ar—	
Spatiotemporal model fit by REML ['sdmTMB']	
Formula: wgtCPUE ~ s(log_depth)	
Mesh: mesh3 (anisotropic covariance)	
Time column: year	
Data: dfr	
Family: delta_gamma(link1 = 'log', link2 = 'log', type = 'poisson-link')	
Delta/hurdle model 1: —————	
Family: binomial(link = 'log')	
coef.est	coef.se
(Intercept)	-5.28 1.19
slog_depth	-0.06 1.58
Smooth terms:	
Std. Dev.	
sds(log_depth)	5.25
Spatiotemporal AR1 correlation (rho): 0.86	
Matérn anisotropic range (spatial): 149.4 to 231.3 at 152 deg.	
Spatial SD: 1.61	
Spatiotemporal marginal AR1 SD: 1.84	
Delta/hurdle model 2: —————	
Family: Gamma(link = 'log')	
coef.est	coef.se
(Intercept)	-1.07 4.40
slog_depth	-0.12 0.13
Smooth terms:	
Std. Dev.	
sds(log_depth)	0
Dispersion parameter: 2.37	
Spatiotemporal AR1 correlation (rho): 1.00	
Matérn anisotropic range (spatial): 841.4 to 1302.7 at 152 deg.	
Spatial SD: 0.00	
Spatiotemporal marginal AR1 SD: 1.34	
REML criterion at convergence: 1103.904	

Table 4. Summary of results for model 'tw'.

###Results for model tw---		
Spatiotemporal model fit by REML ['sdmTMB']		
Formula: wgtCPUE ~ s(log_depth)		
Mesh: mesh3 (anisotropic covariance)		
Time column: year		
Data: dfr		
Family: tweedie(link = 'log')		
coef.est	coef.se	
(Intercept)	-4.43	1.64
slog_depth	0.01	1.69
Smooth terms:		
Std. Dev.		
sds(log_depth)	5.8	
Dispersion parameter: 0.47		
Tweedie p: 1.17		
Matérn anisotropic range (spatial): 148.5 to 195.3 at 136 deg.		
Spatial SD: 2.43		
Spatiotemporal RW SD: 1.14		
REML criterion at convergence: 1123.669		

Table 5. Summary of results for model 'tw-ar'.

### Results for model tw-ar—	
Spatiotemporal model fit by REML ['sdmTMB']	
Formula: wgtCPUE ~ s(log_depth)	
Mesh: mesh3 (anisotropic covariance)	
Time column: year	
Data: dfr	
Family: tweedie(link = 'log')	
coef.est	coef.se
(Intercept)	-6.87 0.90
slog_depth	-0.08 1.82
Smooth terms:	
Std. Dev.	
sds(log_depth)	6.31
Dispersion parameter: 0.37	
Tweedie p: 1.12	
Spatiotemporal AR1 correlation (rho): 0.76	
Matérn anisotropic range (spatial): 115.2 to 168.4 at 139 deg.	
Spatial SD: 1.54	
Spatiotemporal marginal AR1 SD: 2.16	
REML criterion at convergence: 1087.056	

Table 6. AIC scores for the four sdmTMB models.

model	AIC
m3	2259.421
m3-ar	2241.808
tw	2267.339
tw-ar	2196.112

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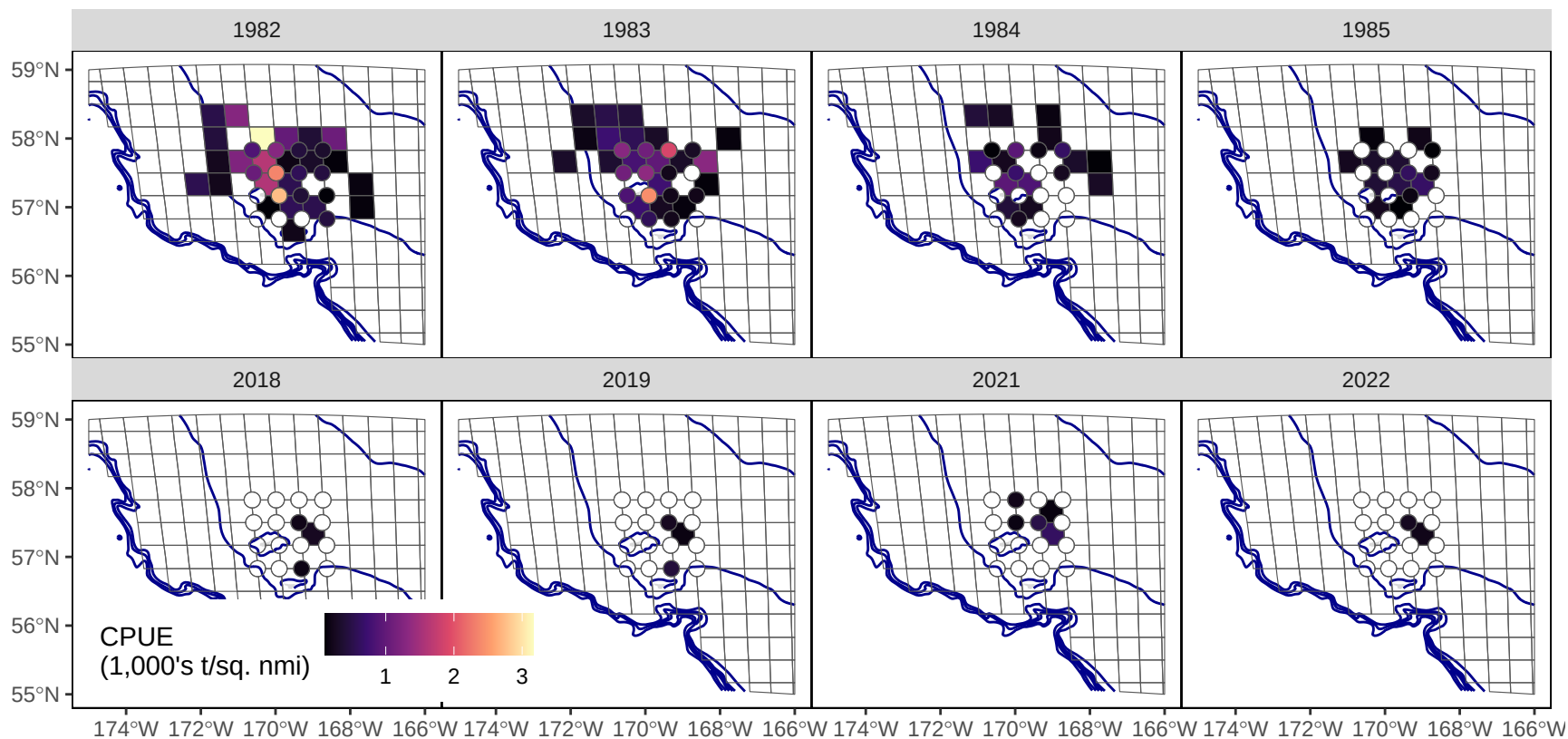


Figure 1. CPUE (1,000's t/sq. nmi.) for mature male PIBKC biomass from the NMFS EBS bottom trawl survey for selected years, illustrating the range contraction and decline in population size over time. The survey has sampled PIBKC annually since 1975 (with the exception of 2020). No mature males were caught in 2023 or 2024.

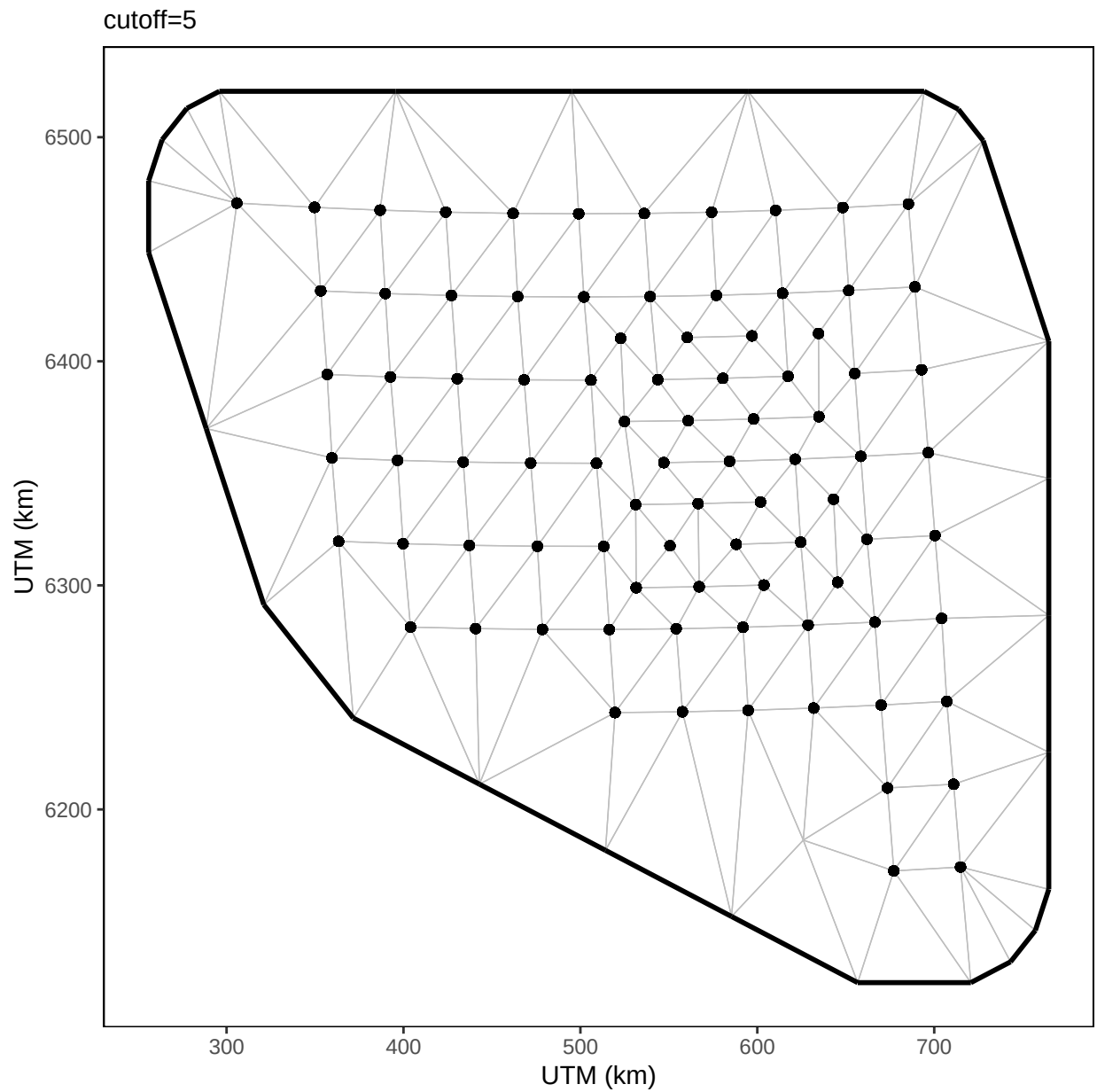


Figure 2. The spatial mesh generated by sdmTMB and used for the sdmTMB models. The points indicate the NMFS EBS bottom trawl survey sampling locations used to create the mesh.

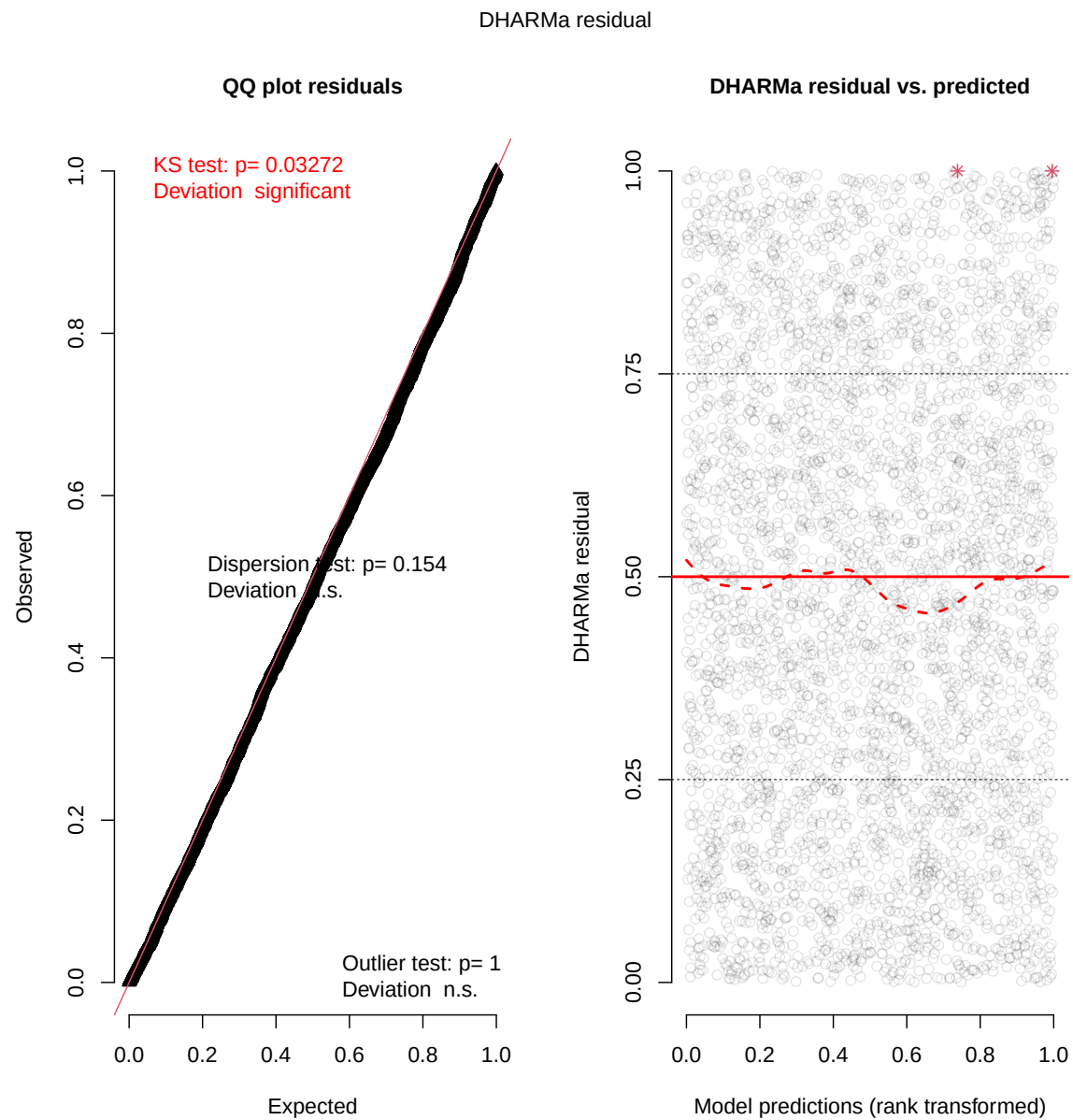


Figure 3. DHAMRa residuals diagnostics plots for 'm3'.

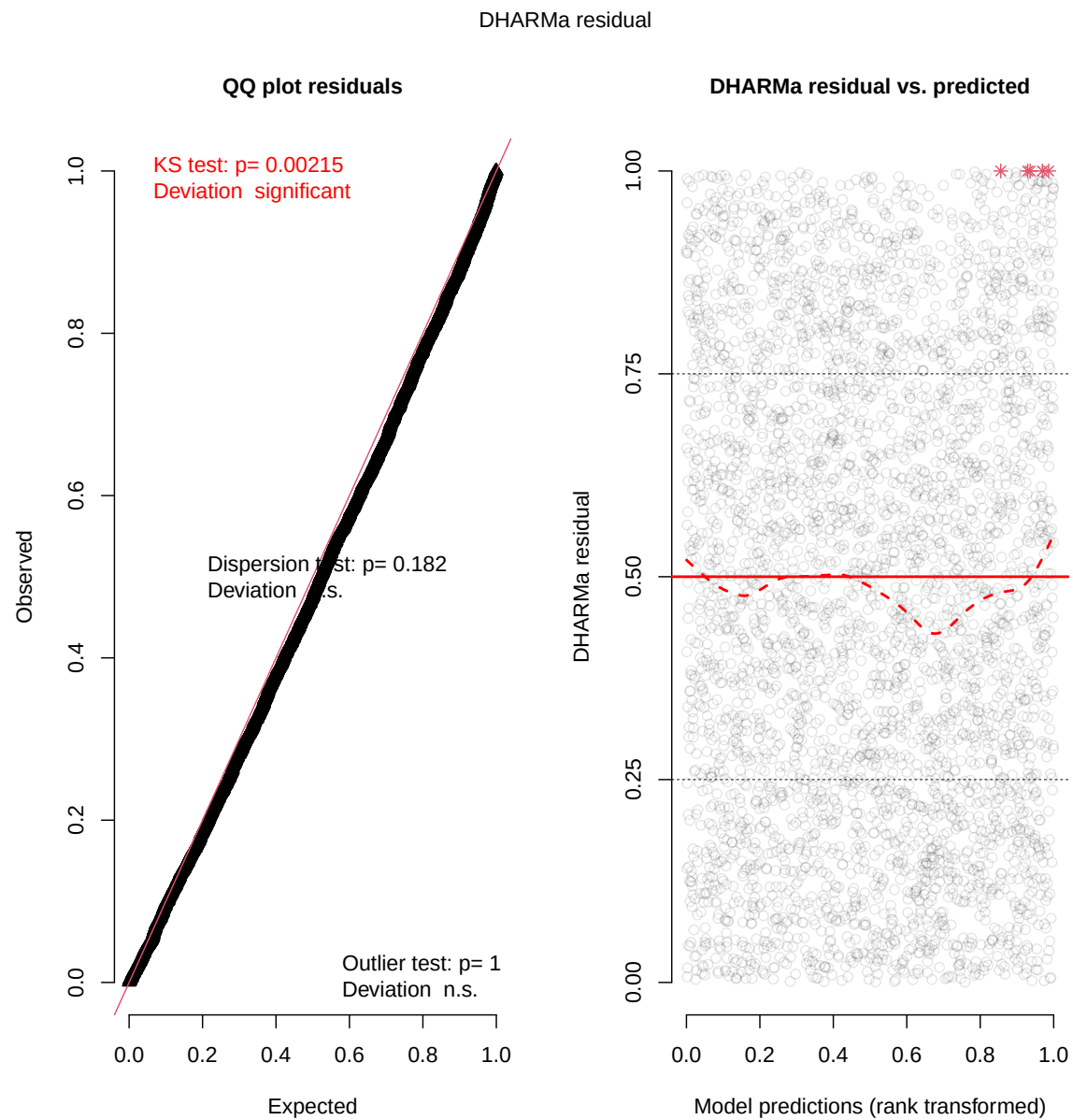


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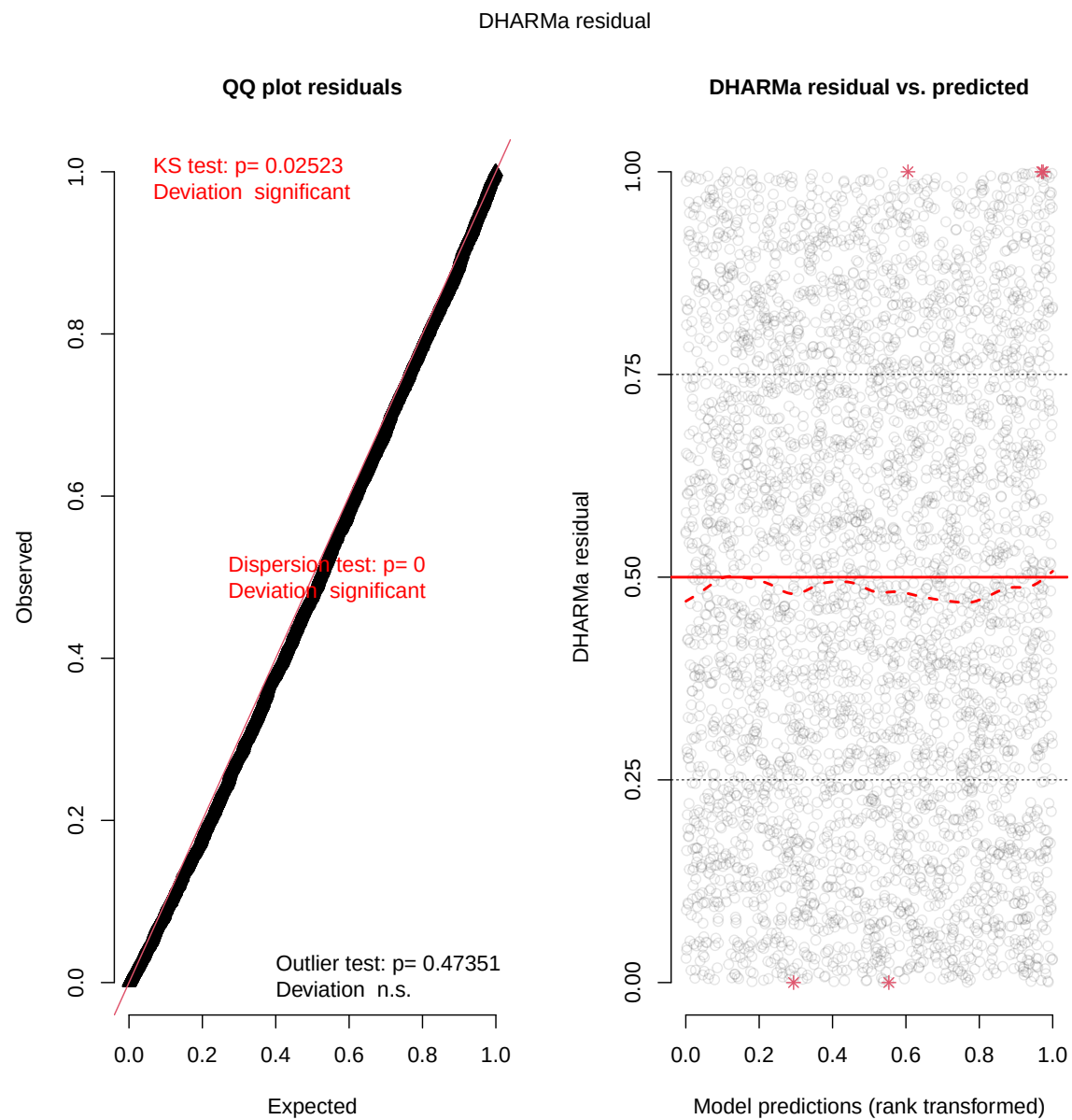


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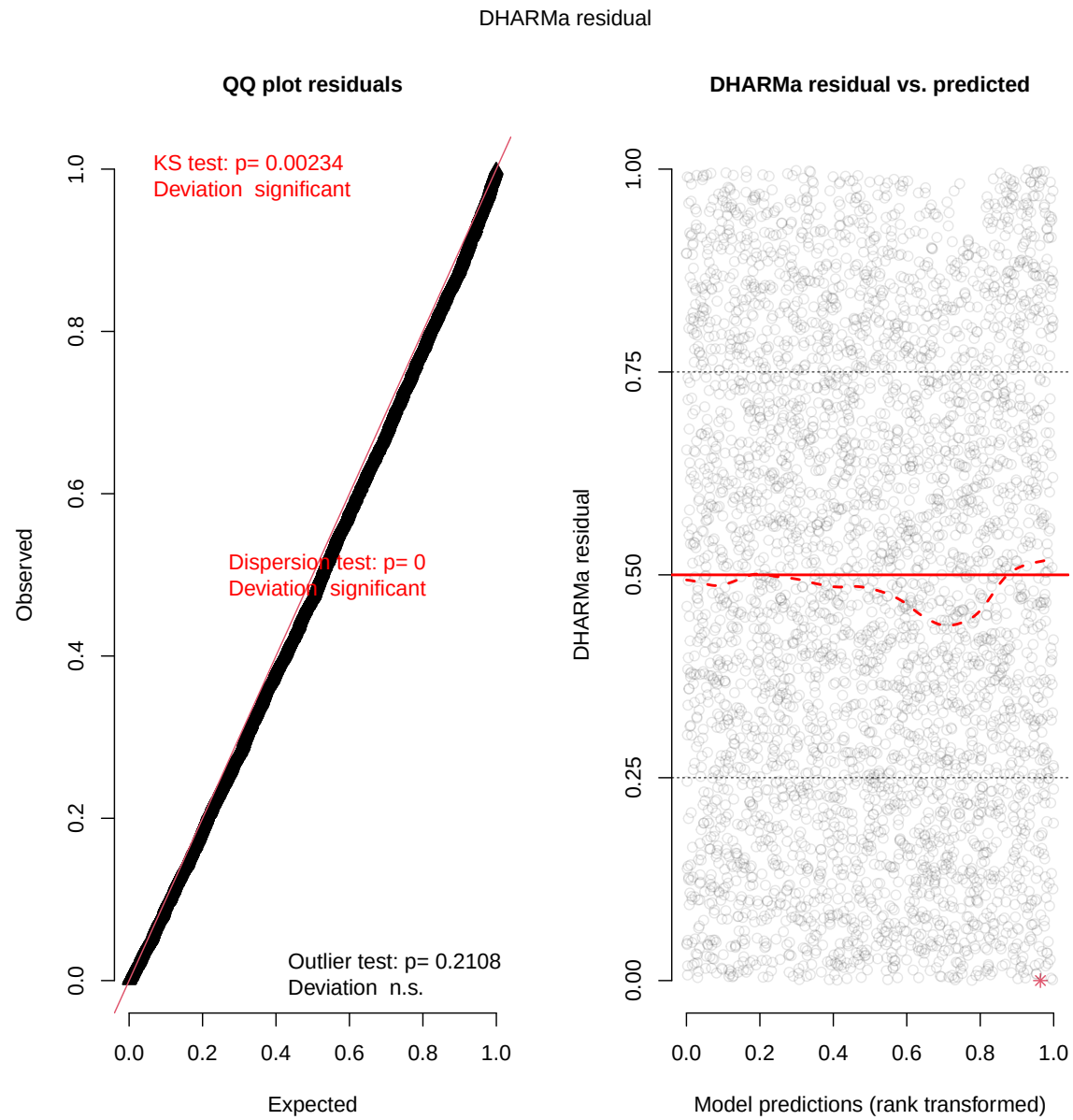


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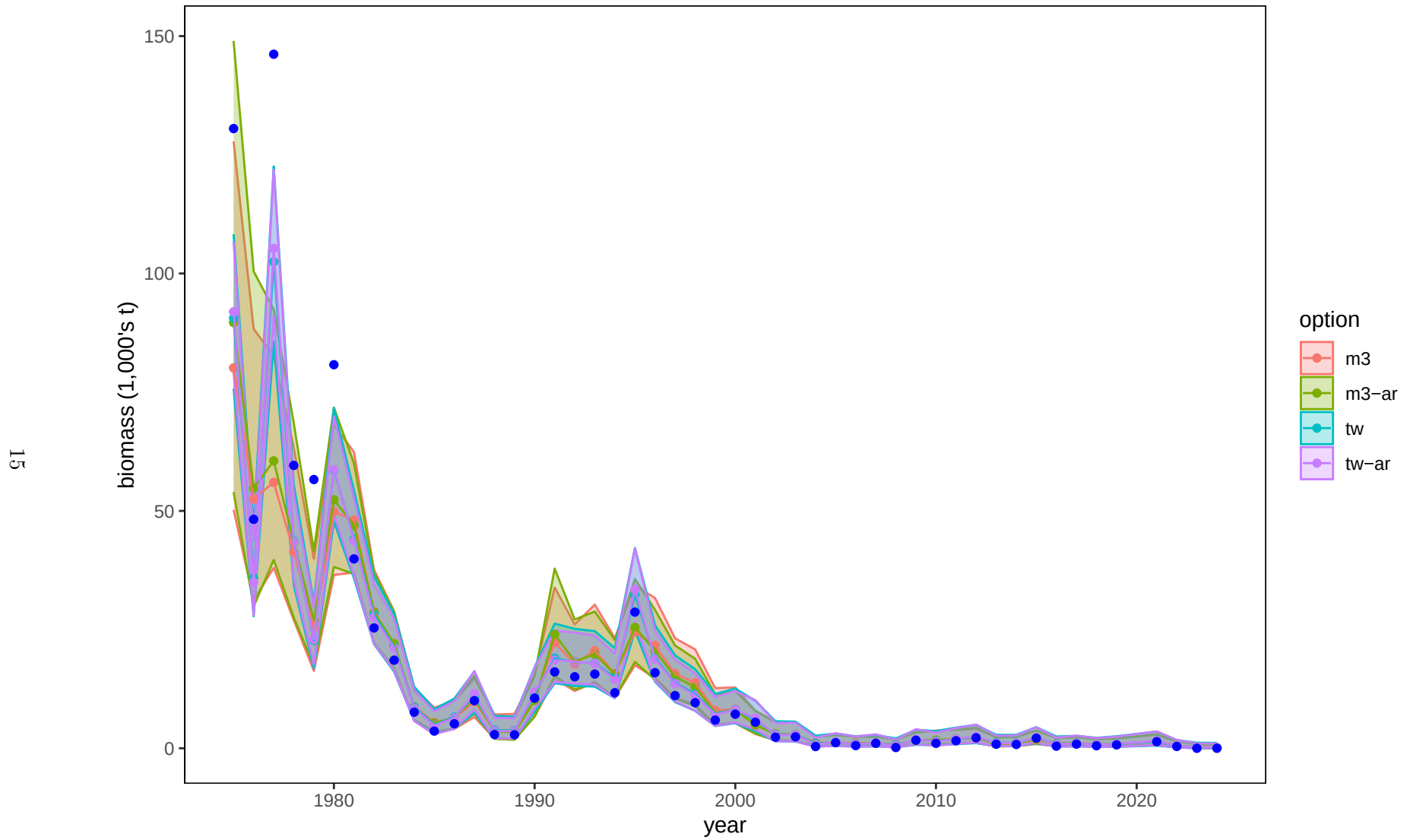


Figure 7. Biomass time series for mature male PIBKC, as estimated by each of the four sdmTMB models. Blue dots indicate the estimates obtained using design-based calculations.

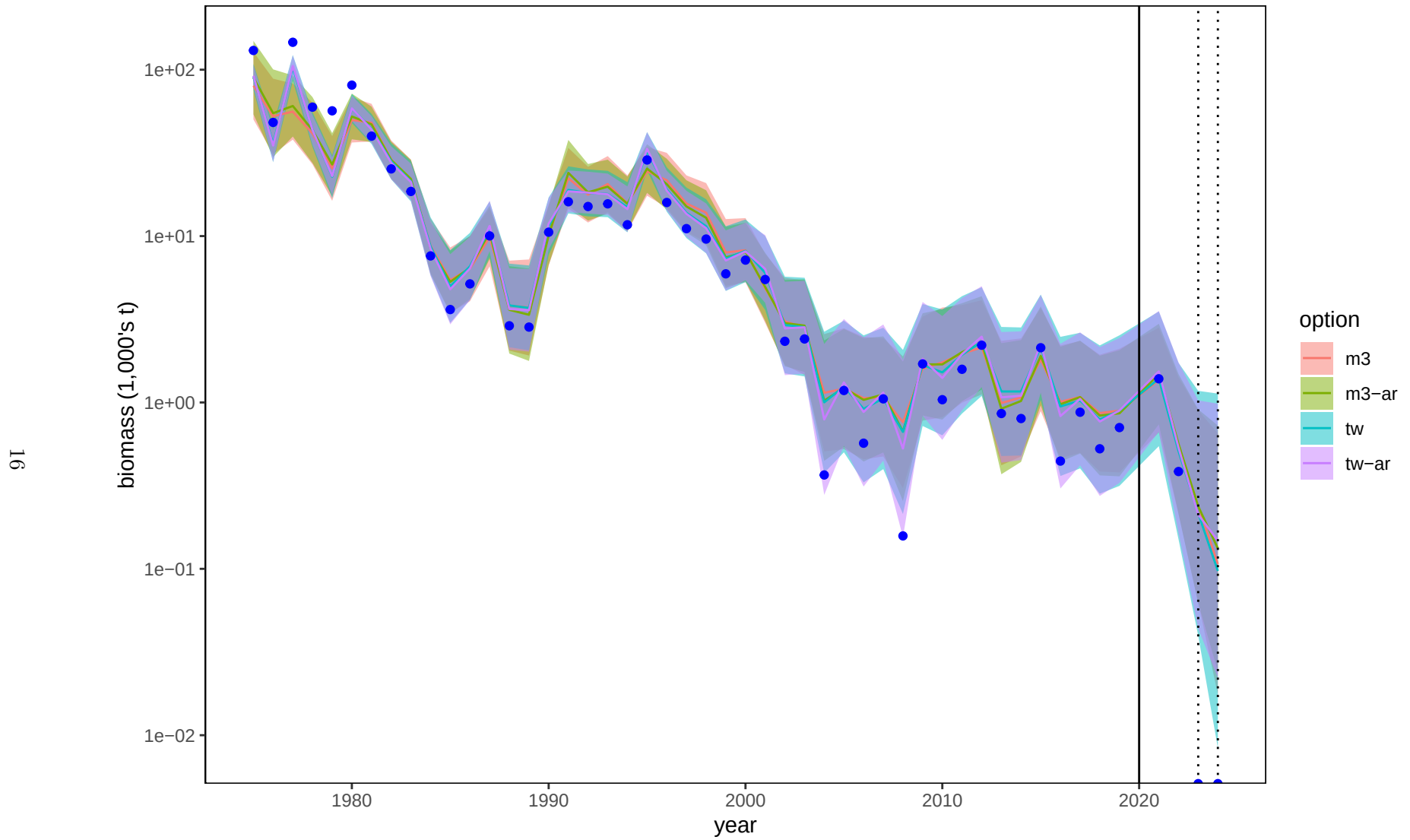


Figure 8. Biomass time series on a log₁₀-scale for mature male PIBKC, as estimated by each of the four sdmTMB models. Blue dots indicate the estimates obtained using design-based calculations.

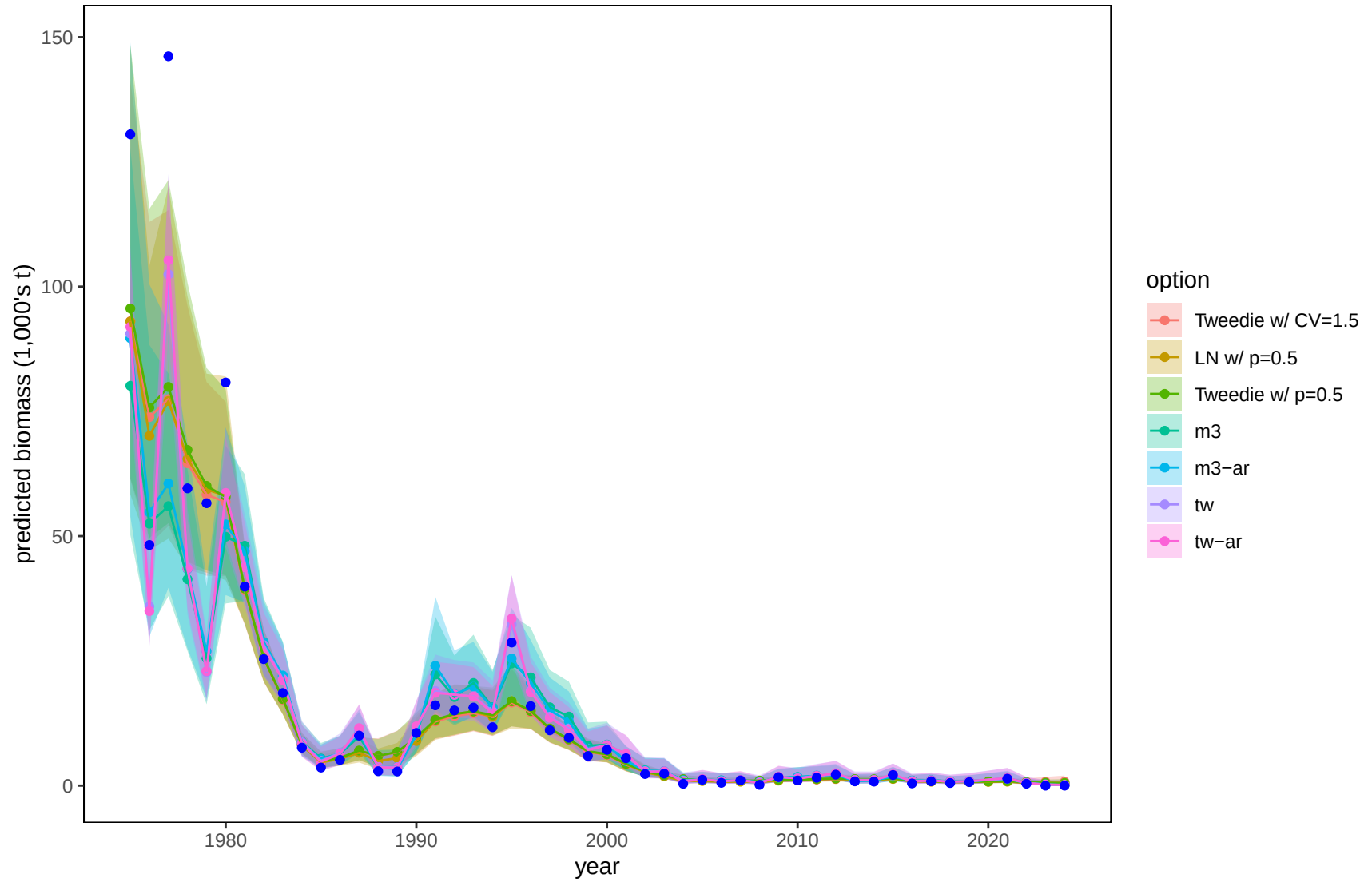


Figure 9. Biomass time series for mature male PIBKC, as estimated using each of the **rema** zeros options and the four **sdmTMB** models. Blue dots indicate the estimates obtained using design-based calculations.

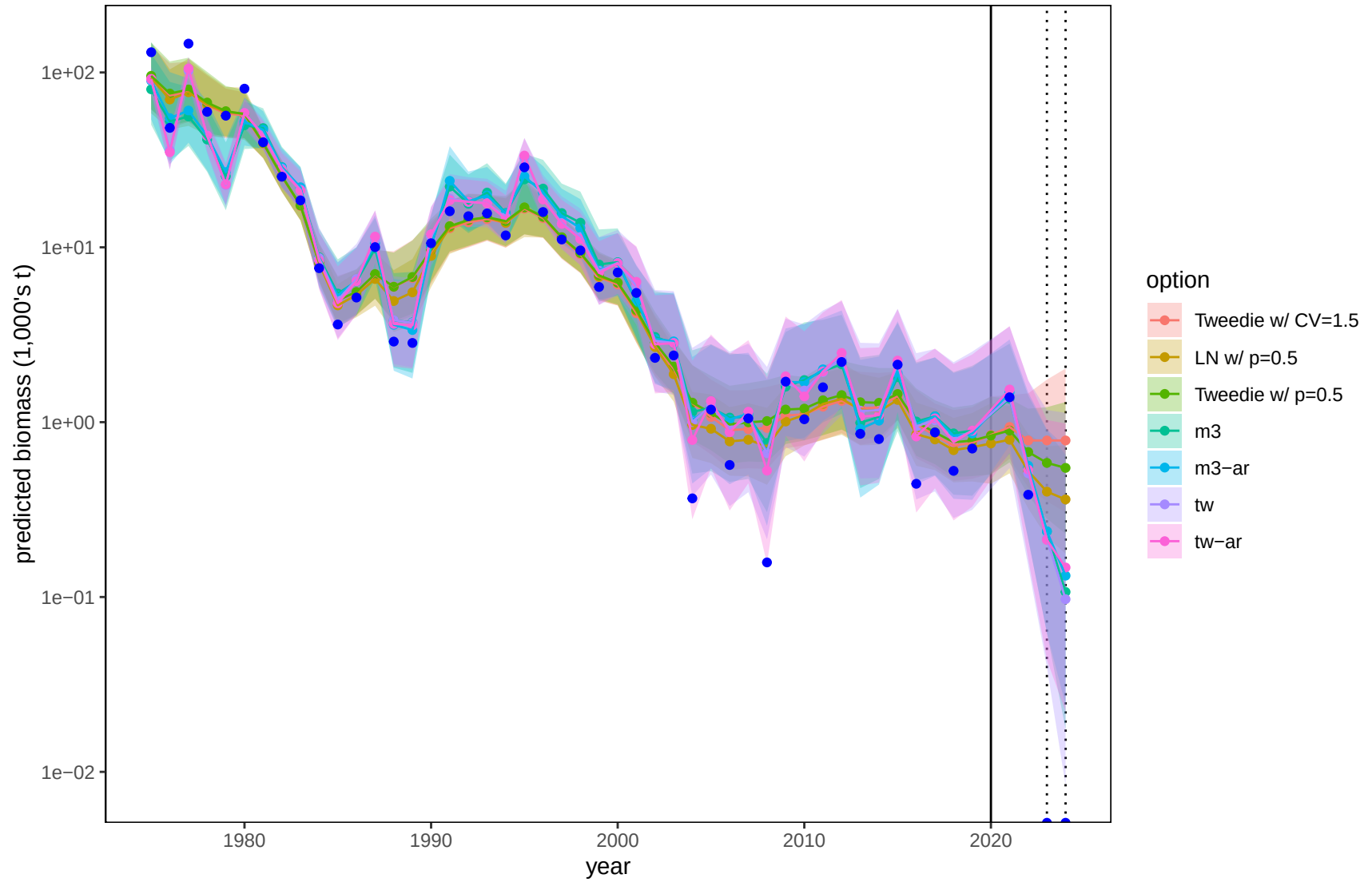


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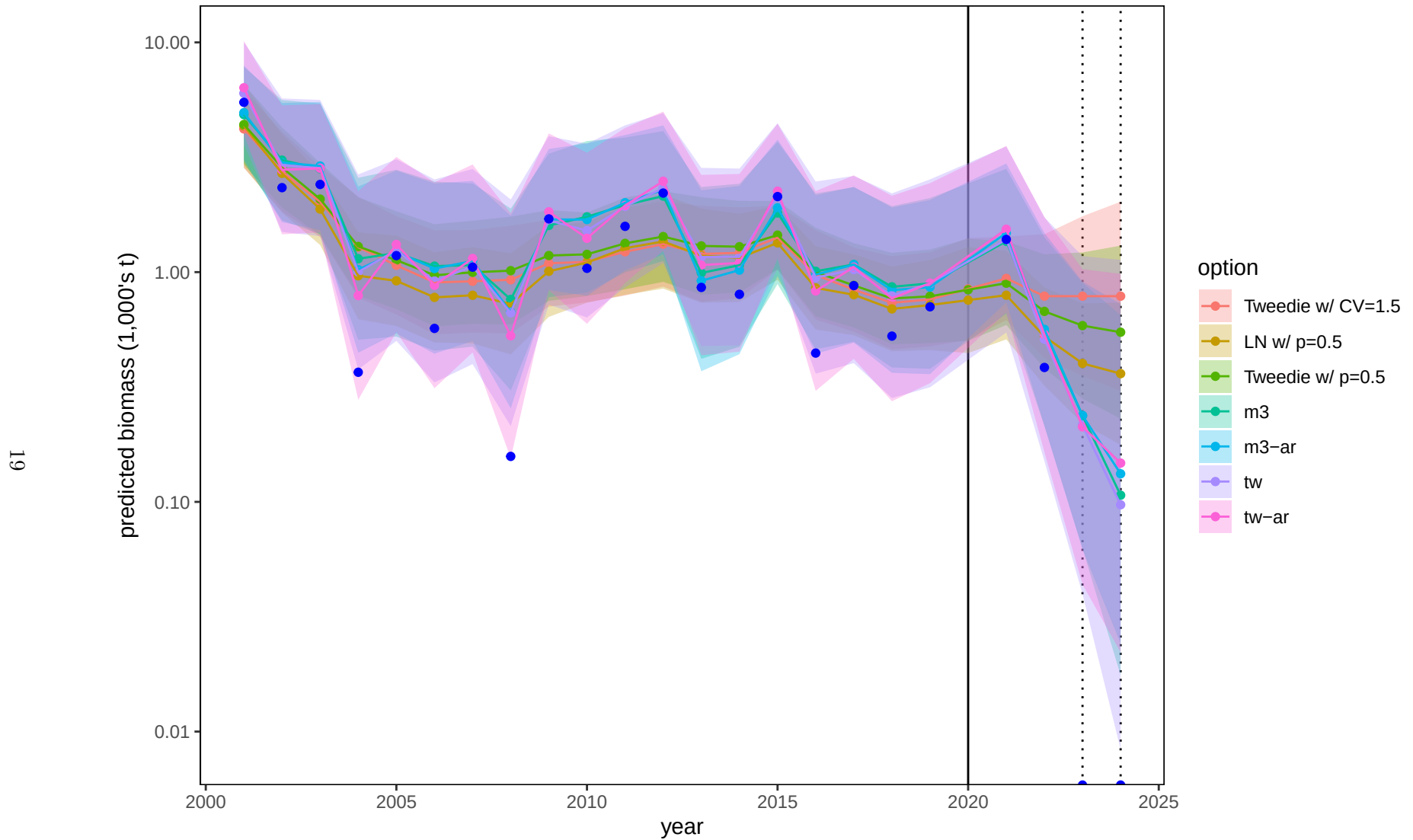


Figure 11. Biomass time series since 2020 for mature male PIBKC on the log10-scale, as estimated using each of the **rema** zeros options and the four sdmTMB models. Blue dots indicate the estimates obtained using design-based calculations.

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